1783: Chloracidobacterium thermophilum Midnight Medium (CTM-M

Final pH: 7
Final volume: 1000 ml

1. The CTM-Medium consists of two parts. Part one is autoclavable and called CTM-Medium basis. Part two is named mixed solution and contains non-autoclavable components. Solid CTM-Medium contains 1% three times washed Bactoagar and is added to the CTM-Medium basis prior autoclaving.

2. Note: Bacto peptone can be omitted with only using amino acids as carbon source, which will maker it a defined medium. Amount of amino acids will then have to be increased to 100mg/L each.

Solution 1	20.000	ml
Solution 2	3.000	ml
Solution 3	2.000	ml
Solution 4	2.500	ml
HEPES buffer	2.400	g

3. Make up to 970 ml with distilled water.

4. Adjust pH to 7.0 with NaOH (2M).

5. Autoclave at 121 °C for 20 min.

6. Seal medium bottle immediately after autoclaving in order to avoid oxygenation.

7. Cool down to 70 °C (in a water bath).

8. Add mixed solution to the CTM basis for completing the CTM-Medium (30 mL),

9. Note: Always prepare freshly! Mix the following components (e.g., in a volume of approx. 30 mL) and add the filter sterilized mixture to the autoclaved CTM-Medium basis:

$NaHCO_3$ (or 10 mL of a 6.25% solution in water)	0.625	g
Na-thioglycolate (or use 1 mL of a 12.5% solution in water, autoclaved under a N_2 gasphase)	0.125	g
Solution 5	0.500	ml
Solution 6	20.000	ml
Vitamin solution A (1000 x stock)	0.250	ml
Vitamin solution B (1000 x stock)	0.250	ml

10. Keep CTM-Medium tightly sealed and between 50-70˚C prior usage.

11. It is strongly recommended to use up the CTM-Medium as soon as possible after preparation! Short term storage of CTM-Medium basis possible under strictly anaerobic conditions. Mixed solutions should then be added freshly before use.

12. Pour CTM-Medium into flasks, bottles or similar and leave a quarter air headspace and seal them.

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Solution 1

1. This is a 50x stock stock solution.

$MgSO_4$ x 7 H_2O	3.75	g
$CaCl_2$ x 2 H_2O	1.80	g
Citric acid	0.30	g
Na-EDTA x 2 H_2O (pH 8, 0,25 M)	0.60	ml
Trace metal solution	50.00	ml

2. Make up to 1000 ml with distilled water.

3. Autoclave at 121 °C for 20 min.

Trace metal solution

H_3BO_3	2.860	g
$MnCl_2$ x 4 H_2O	1.810	g
$ZnSO_4$ x 7 H_2O	0.222	g
Na_2MoO_4 x 2 H_2O	0.390	g
$CuSO_4$ x 5 H_2O	0.079	g
$Co(NO_3)_2$ x 6 H_2O	0.049	g
Distilled water	1000.000	ml

Solution 2

K_2HPO_4	15.30	

1. Make up to 1000 ml with distilled water.

2. Autoclave at 121 °C for 20 min.

Solution 3

Ferric ammonium citrate	12.00	g

1. Make up to 1000 ml with distilled water.

2. Autoclave at 121 °C for 20 min.

Solution 4

2-Oxoglutarate	14.60	g

1. Make up to 1000 ml with distilled water.

2. Autoclave at 121 °C for 20 min.

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Vitamin solution A (1000 x stock) (from medium 1783)

Biotin	10.00	mg
Riboflavin	10.00	mg
Thiamine HCl	100.00	mg
Thiamine pyrophosphate	100.00	mg
L-Ascorbic acid	100.00	mg
Calcium pantothenate	100.00	mg
Folic acid	100.00	mg
Nicotinamide	100.00	mg
Nicotinic acid	100.00	mg
p-Aminobenzoic acid	100.00	mg
Pyridoxine hydrochloride	100.00	mg
Lipoic acid	100.00	mg
beta-NAD	100.00	mg
Potassium phosphate buffer (pH 7.2, 10 mM)	100.00	ml

1. Dissolve the vitamins in 10mM phosphate buffer, pH7.2.

2. Titrate with NaOH until vitamins are dissolved, filter sterilize afterwards, and freeze in appropriate aliquots until usage.

Vitamin solution B (1000 x stock) (from medium 1783)

Vitamin B_{12}	100.00	mg
Distilled water	100.00	ml

1. Dissolve the vitamin B_{12} in water.

2. Titrate with HCl until vitamin B_{12} is dissolved, filter sterilize afterwards, and freeze in appropriate aliquots until usage.

Solution 5

Bacto peptone (BD-Difco)	10.00	g
Distilled water	100.00	ml

Autoclave at 121 °C for 20 min.

Solution 6

L-Alanine	100.00	mg
L-Arginine	100.00	mg
L-Asparagine	100.00	mg
L-Aspartic acid	100.00	mg
L-Cysteine	100.00	mg
L-Glutamic acid	100.00	mg

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L-Glutamine	100.00	mg
Glycine	100.00	mg
L-Histidine	100.00	mg
L-Isoleucine	100.00	mg
L-Leucine	100.00	mg
L-Lysine	100.00	mg
L-Methionine	100.00	mg
L-Phenylalanine	100.00	mg
L-Proline	100.00	mg
L-Serine	100.00	mg
L-Threonine	100.00	mg
L-Tryptophan	100.00	mg
L-Tyrosine	100.00	mg
L-Valine	100.00	mg
Distilled water	400.00	ml

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